

## TUCSON 11 W, AZ, USA

WMO: 722750

Lat: 32.2395N

Lon: 111.1696W

Elev: 2733

StdP: 13.30

Time Zone: -7.00 (W07)

Period: 02-19

WBAN: 53131

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |      | Humidification DP/MCDB and HR |     |      |     |     |      | Coldest Month WS/MCDB |      |     |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|------|-------------------------------|-----|------|-----|-----|------|-----------------------|------|-----|------|-----------------------|------|-------|
|               |            |      | 99.6%                         |     |      | 99% |     |      | 0.4%                  |      | 1%  |      |                       |      |       |
|               | 99.6%      | 99%  | DP                            | HR  | MCDB | DP  | HR  | MCDB | WS                    | MCDB | WS  | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c)  | (d)                           | (e) | (f)  | (g) | (h) | (i)  | (j)                   | (k)  | (l) | (m)  | (n)                   | (o)  | (p)   |
| 12            | 32.3       | 35.3 | -2.8                          | 5.2 | 67.9 | 1.2 | 6.5 | 68.8 | 11.3                  | 55.7 | 9.8 | 56.5 | 2.0                   | N/A  | 0.360 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |       |      |       |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|-------|------|-------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%    |      | 2%    |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB    | MCWB | DB    | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)   | (f)  | (g)   | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 6             | 28.2                   | 104.5           | 64.1 | 102.5 | 64.0 | 100.4 | 63.8 | 71.8                | 85.5 | 71.0 | 85.5 | 70.2 | 85.7 | 5.3                  | N/A  |

| Dehumidification DP/MCDB and HR |       |      |      |       |      |      |       |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|-------|------|------|-------|------|------|-------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |       |      | 1%   |       |      | 2%   |       |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR    | MCDB | DP   | HR    | MCDB | DP   | HR    | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)   | (c)  | (d)  | (e)   | (f)  | (g)  | (h)   | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  | (p)               |
| 68.8                            | 117.2 | 75.9 | 67.5 | 111.8 | 76.3 | 66.0 | 106.0 | 77.3 | 37.4          | 85.2 | 36.6 | 84.9 | 35.9 | 84.9 | 76.8              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |     | Extreme Annual Temperature |      |                    |     | n-Year Return Period Values of Extreme Temperature |      |            |      |            |      |            |      |       |     |
|-------------------|------|-----|----------------------------|------|--------------------|-----|----------------------------------------------------|------|------------|------|------------|------|------------|------|-------|-----|
|                   |      |     | Mean                       |      | Standard Deviation |     | n=5 years                                          |      | n=10 years |      | n=20 years |      | n=50 years |      |       |     |
| 1%                | 2.5% | 5%  | Min                        | Max  | Min                | Max | Min                                                | Max  | Min        | Max  | Min        | Max  | Min        | Max  |       |     |
| (a)               | (b)  | (c) | (e)                        | (f)  | (g)                | (h) | (i)                                                | (j)  | (k)        | (l)  | (m)        | (n)  | (o)        | (p)  |       |     |
| 9.4               | 8.4  | 7.3 | DB                         | 25.8 | 108.5              | 3.6 | 2.0                                                | 23.2 | 110.0      | 21.1 | 111.2      | 19.1 | 112.3      | 16.5 | 113.8 | (4) |
|                   |      |     | WB                         | 22.0 | 73.9               | 5.1 | 1.3                                                | 18.3 | 74.8       | 15.4 | 75.6       | 12.5 | 76.3       | 8.8  | 77.2  | (5) |

## Monthly Climatic Design Conditions

|                                                                                  |             | Annual | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec  |
|----------------------------------------------------------------------------------|-------------|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|------|
|                                                                                  |             | (d)    | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)  |
| Temperatures,<br>Degree-Days<br>and<br>Degree-Hours                              | DBAvg       | 70.2   | 53.8  | 55.3  | 62.1  | 67.7  | 75.4  | 85.9  | 87.2  | 85.3  | 81.6  | 72.0  | 62.1  | 52.9 |
|                                                                                  | DBStd       | 13.87  | 6.82  | 7.19  | 7.12  | 6.52  | 6.52  | 4.67  | 4.35  | 4.08  | 4.69  | 6.53  | 7.40  | 7.01 |
|                                                                                  | HDD50       | 134    | 40    | 29    | 5     | 1     | 0     | 0     | 0     | 0     | 0     | 0     | 7     | 52   |
|                                                                                  | HDD65       | 1350   | 350   | 276   | 139   | 46    | 6     | 0     | 0     | 0     | 0     | 19    | 138   | 376  |
|                                                                                  | CDD50       | 7504   | 158   | 178   | 380   | 532   | 789   | 1076  | 1153  | 1095  | 948   | 683   | 369   | 143  |
|                                                                                  | CDD65       | 3245   | 2     | 5     | 49    | 128   | 330   | 626   | 688   | 630   | 498   | 237   | 50    | 2    |
|                                                                                  | CDH74       | 45269  | 33    | 143   | 798   | 1857  | 4751  | 9949  | 9805  | 8450  | 6234  | 2653  | 561   | 35   |
|                                                                                  | CDH80       | 25383  | 0     | 17    | 207   | 692   | 2522  | 6468  | 5966  | 4897  | 3382  | 1108  | 124   | 0    |
| Wind                                                                             | WSAvg       | 3.5    | 3.3   | 3.3   | 3.6   | 3.9   | 4.0   | 3.8   | 3.4   | 3.0   | 3.3   | 3.4   | 3.5   | 3.2  |
| Precipitation                                                                    | PrecAvg     | 10.7   | 1.0   | 0.9   | 0.8   | 0.3   | 0.2   | 0.2   | 1.9   | 2.0   | 1.1   | 0.6   | 0.6   | 1.1  |
|                                                                                  | PrecMax     | 18.5   | 5.2   | 3.7   | 3.1   | 1.3   | 1.2   | 1.9   | 5.1   | 5.0   | 2.7   | 3.8   | 2.0   | 3.5  |
|                                                                                  | PrecMin     | 5.7    | 0.0   | 0.0   | 0.0   | 0.0   | 0.0   | 0.0   | 0.2   | 0.6   | 0.2   | 0.0   | 0.0   | 0.0  |
|                                                                                  | PrecStd     | 3.0    | 1.2   | 0.9   | 0.7   | 0.3   | 0.3   | 0.4   | 1.1   | 1.0   | 0.6   | 0.7   | 0.5   | 1.1  |
| Monthly Design<br>Dry Bulb<br>and<br>Mean Coincident<br>Wet Bulb<br>Temperatures | 0.4%        | DB     | 77.6  | 82.5  | 89.3  | 93.6  | 100.9 | 108.1 | 107.1 | 105.2 | 101.5 | 95.4  | 87.2  | 77.8 |
|                                                                                  |             | MCWB   | 49.9  | 51.5  | 53.9  | 56.2  | 58.8  | 62.9  | 66.2  | 66.9  | 64.4  | 60.3  | 54.0  | 50.6 |
|                                                                                  | 2%          | DB     | 74.2  | 78.6  | 85.2  | 90.2  | 97.6  | 104.7 | 104.3 | 102.4 | 98.9  | 92.2  | 83.7  | 74.2 |
|                                                                                  |             | MCWB   | 49.5  | 50.4  | 52.8  | 54.8  | 57.6  | 62.0  | 66.1  | 67.4  | 64.5  | 59.0  | 54.3  | 49.9 |
|                                                                                  | 5%          | DB     | 70.8  | 74.6  | 81.9  | 87.2  | 95.1  | 102.8 | 102.0 | 100.0 | 96.8  | 89.4  | 80.3  | 70.3 |
|                                                                                  |             | MCWB   | 48.0  | 49.4  | 51.4  | 53.7  | 57.0  | 61.5  | 66.2  | 66.8  | 64.2  | 58.7  | 53.7  | 49.1 |
|                                                                                  | 10%         | DB     | 67.4  | 70.5  | 78.6  | 84.2  | 92.4  | 100.7 | 99.5  | 97.5  | 94.3  | 86.6  | 77.0  | 66.6 |
|                                                                                  |             | MCWB   | 46.8  | 48.4  | 50.6  | 52.8  | 55.8  | 60.9  | 66.3  | 66.8  | 64.1  | 57.7  | 52.8  | 47.9 |
| Monthly Design<br>Wet Bulb<br>and<br>Mean Coincident<br>Dry Bulb<br>Temperatures | 0.4%        | WB     | 55.3  | 56.1  | 55.4  | 58.0  | 60.4  | 69.3  | 73.0  | 73.3  | 71.2  | 67.8  | 58.8  | 55.3 |
|                                                                                  |             | MCDB   | 65.1  | 63.2  | 78.3  | 87.6  | 91.1  | 87.6  | 84.5  | 86.9  | 83.7  | 73.8  | 75.5  | 65.7 |
|                                                                                  | 2%          | WB     | 52.9  | 54.0  | 54.0  | 56.4  | 59.0  | 67.7  | 71.9  | 72.1  | 70.0  | 64.5  | 56.9  | 53.1 |
|                                                                                  |             | MCDB   | 63.7  | 65.7  | 76.8  | 86.0  | 92.5  | 88.0  | 84.9  | 86.3  | 83.3  | 79.1  | 74.2  | 66.6 |
|                                                                                  | 5%          | WB     | 51.0  | 52.0  | 52.6  | 55.1  | 57.4  | 66.1  | 71.1  | 71.2  | 69.1  | 62.5  | 55.4  | 51.4 |
|                                                                                  |             | MCDB   | 63.9  | 68.3  | 75.7  | 83.7  | 90.7  | 90.1  | 85.2  | 86.2  | 83.1  | 80.1  | 73.9  | 65.0 |
|                                                                                  | 10%         | WB     | 49.2  | 50.5  | 51.4  | 53.9  | 56.3  | 64.5  | 70.3  | 70.3  | 68.2  | 60.8  | 54.0  | 49.9 |
|                                                                                  |             | MCDB   | 63.8  | 68.7  | 74.7  | 81.9  | 88.2  | 92.6  | 85.6  | 86.4  | 82.9  | 79.5  | 73.2  | 63.4 |
| Mean Daily<br>Temperature<br>Range                                               | 5% DB       | MDBR   | 22.8  | 23.9  | 26.2  | 27.6  | 28.2  | 28.2  | 22.4  | 22.4  | 23.1  | 24.9  | 24.2  | 22.2 |
|                                                                                  |             | MCDBR  | 26.8  | 30.0  | 30.6  | 31.1  | 30.9  | 29.4  | 25.1  | 25.7  | 25.8  | 27.1  | 27.6  | 26.7 |
|                                                                                  |             | MCWBR  | 12.2  | 12.8  | 12.1  | 12.1  | 11.1  | 9.4   | 6.3   | 6.5   | 7.5   | 9.7   | 11.0  | 11.8 |
|                                                                                  | 5% WB       | MCDBR  | 20.8  | 24.5  | 27.0  | 29.0  | 29.4  | 23.5  | 20.5  | 21.0  | 20.3  | 21.2  | 23.8  | 22.0 |
|                                                                                  |             | MCWBR  | 10.4  | 10.9  | 10.6  | 11.0  | 10.3  | 7.4   | 4.8   | 5.4   | 6.2   | 8.0   | 10.2  | 10.4 |
| Clear-Sky<br>Solar<br>Irradiance                                                 | taub        | 0.260  | 0.264 | 0.283 | 0.296 | 0.306 | 0.327 | 0.385 | 0.372 | 0.332 | 0.295 | 0.269 | 0.259 |      |
|                                                                                  | taud        | 2.636  | 2.628 | 2.541 | 2.487 | 2.492 | 2.440 | 2.409 | 2.434 | 2.516 | 2.571 | 2.603 | 2.631 |      |
|                                                                                  | Ebn at Noon | 312    | 320   | 318   | 315   | 310   | 302   | 284   | 287   | 297   | 304   | 305   | 307   |      |
|                                                                                  | Edh at Noon | 24     | 26    | 31    | 34    | 35    | 36    | 37    | 36    | 32    | 28    | 24    | 23    |      |
| All-Sky Solar<br>Radiation                                                       | RadAvg      | 1139   | 1440  | 1911  | 2349  | 2602  | 2621  | 2200  | 2072  | 1893  | 1605  | 1259  | 1041  |      |
|                                                                                  | RadStd      | 78     | 111   | 90    | 69    | 90    | 133   | 126   | 70    | 87    | 97    | 60    | 47    |      |

## Historical Trends

|                         |       | DBAvg | Heating |        | Cooling |       |       | Degree-Days |       |       |       |
|-------------------------|-------|-------|---------|--------|---------|-------|-------|-------------|-------|-------|-------|
|                         |       |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD50       | HDD65 | CDD50 | CDD65 |
|                         | (d)   | (e)   | (f)     | (g)    | (h)     | (i)   | (j)   | (k)         | (l)   | (m)   |       |
| Station Only            | N/A   | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | +326  | N/A   |       |
| Regional (14 neighbors) | +0.49 | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | -135  | +187  | N/A   |

Nomenclature: See separate page